

Fundamental of AI

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Unit – I

Introduction

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Outline

- History & overview of Artificial Intelligence
- Definition of Artificial Intelligence
- Artificial Narrow Intelligence, Artificial General Intelligence, Artificial Super Intelligence
- Concepts of Production, Agents and Environments
- Characteristic of Intelligent Agents, Concept of Rationality, Nature of Environments.

Introduction to AI

What is AI?

- **Definition:** AI is the simulation of human intelligence in machines that are programmed to think, learn, and problem-solve.
- According to **Gemini:** AI is a collection of technologies that allow computers to perform tasks that humans normally do. AI systems use math and logic to simulate human reasoning and make decisions.
- According to **IBM:** AI is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.

Early Concepts of AI

Ancient Mythology:

- Stories of automatons (mechanical beings) like **Talos** (a man of bronze who protected Crete from pirates and invaders) in Greek mythology.

Philosophers:

- Aristotle and the idea of logical reasoning.
- Early concepts of machine intelligence in the 17th century, like René Descartes and Thomas Hobbes (who used rationalism to explain the physical and mental world).

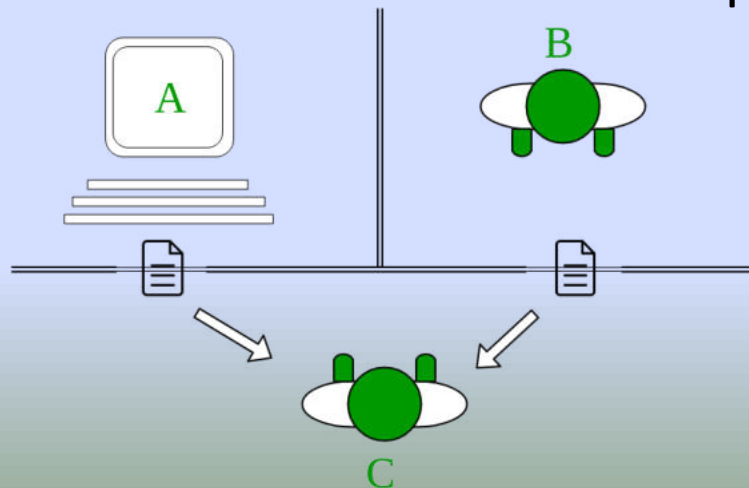
Alan Turing:

- 1936: Developed the **Turing Machine** concept, laying the groundwork for computer science.
- 1950: Published the famous paper, "Computing Machinery and Intelligence", introducing the **Turing Test** for machine intelligence.

Early Concepts of AI

Turing Test: (Acting Humanly)

- The Turing Test is a widely recognized benchmark for evaluating a machine's ability to demonstrate human-like intelligence.
- The **core idea** is simple: A human judge engages in a text-based conversation with both a human and a machine.
- The judge's task is to determine which participant is human and which is the machine. If the judge is unable to distinguish between the human and the machine based solely on the conversation, the machine is said to have passed the Turing Test.



Early Concepts of AI

Thinking humanly: The cognitive modeling approach

- Once we gather enough data, we can create a model to simulate the human process. This model can be used to create software that can think like humans.
- All we care about is the output of the program given a particular input. If the program behaves in a way that matches human behavior, then we can say that humans have a similar thinking mechanism.
- Within computer science, there is a field of study called **Cognitive Modeling** that deals with simulating the human thinking process.
- It tries to understand how humans solve problems. It takes the mental processes that go into this problem solving process and turns it into a software model. This model can then be used to simulate human behavior.
- Cognitive modeling is used in a variety of AI applications such as deep learning, expert systems, Natural Language Processing, robotics, and so on.

Early Concepts of AI

Thinking rationally: The “laws of thought” approach

- **Rationality** refers to doing the right thing in a given circumstance.
- The Greek philosopher Aristotle was one of the first to attempt to codify “right thinking,” that is, irrefutable reasoning processes.
- His syllogisms provided patterns for argument structures that always yielded correct conclusions when given correct premises.
- for example, “Socrates is a man; all men are mortal; therefore, Socrates is mortal.” These laws of thought were supposed to govern the operation of the mind; their study initiated the field called logic.

Early Concepts of AI

Acting rationally: The rational agent approach

- An agent is just something that acts (agent comes from the Latin agere, to do).
- Of course, all computer programs do something, but computer agents are expected to do more: operate autonomously, perceive their environment, persist over a prolonged time period, adapt to change, and create and pursue goals.
- Rational agents need to be performed in such a way that there is maximum benefit to the entity performing the action.
- An agent is said to act rationally if, given a set of rules, it takes actions to achieve its goals.
- It just perceives and acts according to the information that's available. This system is used a lot in AI to design robots when they are sent to navigate unknown terrains.

Early Concepts of AI

Thinking Humanly

“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985)

“[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)

Thinking Rationally

“The study of mental faculties through the use of computational models.”
(Charniak and McDermott, 1985)

“The study of the computations that make it possible to perceive, reason, and act.”
(Winston, 1992)

Acting Humanly

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990)

“The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)

Acting Rationally

“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998)

“AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

The Birth of AI (1950s - 1960s)

1956: Dartmouth Conference:

- The official birth of AI as a field of study.
- Organizers: John McCarthy (a mathematics professor), Marvin Minsky, Nathaniel Rochester, and Claude Shannon.
- Term "Artificial Intelligence" coined by John McCarthy.

Early Programs:

- **Logic Theorist** (1955): Developed by Allen Newell and Herbert A. Simon, the first AI program that could prove mathematical theorems.
- **General Problem Solver** (1959): Another Newell-Simon development, aiming to simulate human problem-solving.

Growth and Setbacks (1960s - 1970s)

ELIZA (1966):

- A natural language processing program created by Joseph Weizenbaum that simulated conversation.

Expert Systems:

- AI programs designed to solve specific problems in specialized domains
- Example: medical diagnosis done by **MYCIN** in the 1970s.

The AI Winter (1970s):

- Limited progress led to disillusionment and reduced funding for AI research, known as the "AI Winter."

AI Resurgence (1980s - 1990s)

Expert Systems and Neural Networks:

- Expert systems became more widely used in business and medicine.
- **Backpropagation:** Rediscovery of the neural network algorithm, spurring advancements in machine learning.

Deep Blue:

- 1997: IBM's Deep Blue defeated world chess champion Garry Kasparov, showcasing AI's ability to handle complex tasks.



The Rise of Machine Learning and Big Data (2000s - 2010s)

Machine Learning:

- AI systems began to "learn" from data, improving their performance over time.

Big Data and Computing Power:

- Growth of data and increased computing power enabled more sophisticated AI models.

Breakthroughs in Speech Recognition and Image Processing:

- Voice assistants like Siri (2011), Google Now (2012), and Amazon Alexa emerged.
- **AlexNet:** an Image Classification model that transformed deep learning. It was introduced by Geoffrey Hinton and his team in 2012, and marked a key event in the history of deep learning, showcasing the strengths of CNN architectures and its vast applications

Modern AI (2020s and Beyond)

AI in Everyday Life:

- AI-powered tools like self-driving cars, AI assistants, and recommendation systems.

Generative AI:

- Tools like GPT (Generative Pretrained Transformer) and DALL·E for generating text and images.

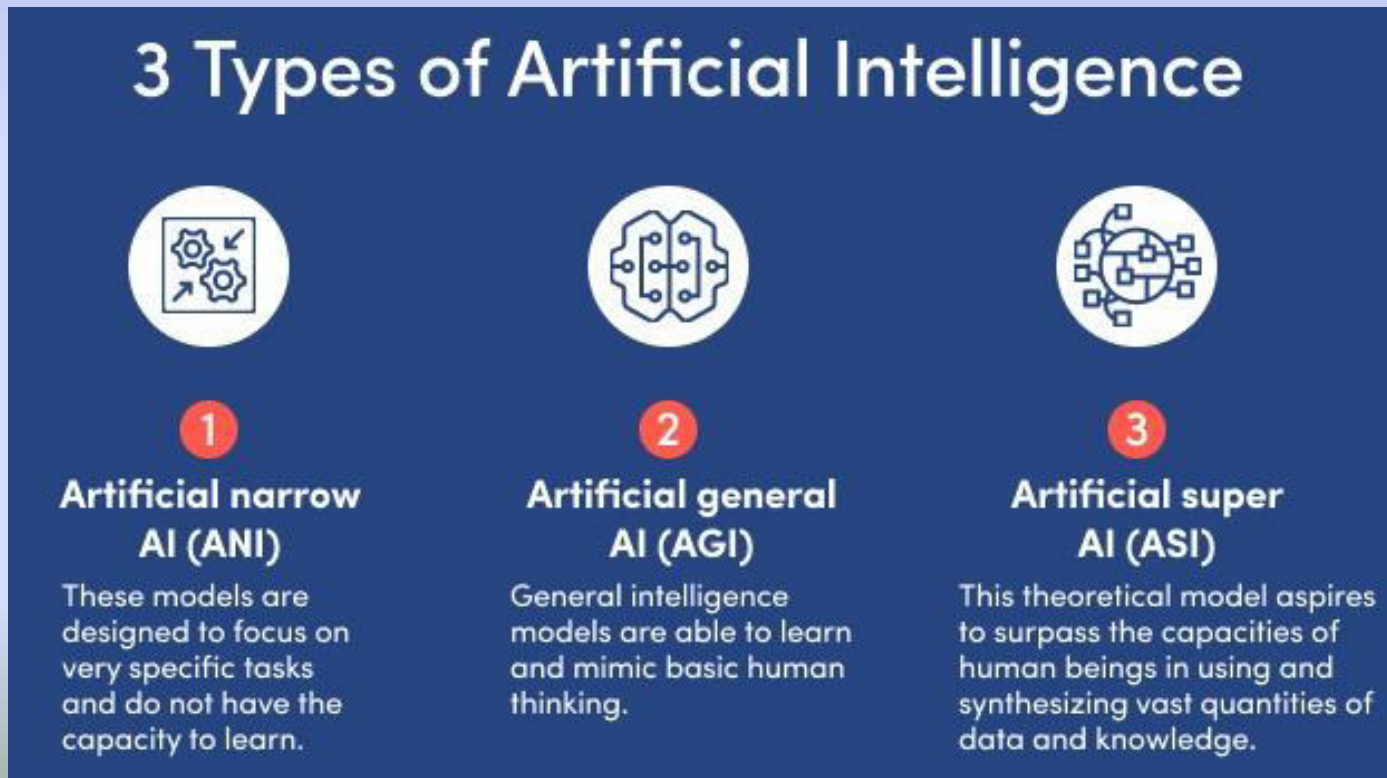
AI Ethics and Governance:

- Growing discussions on the ethical implications of AI, privacy concerns, and regulation.

Categories of AI

Categories of AI:

- ANI (Artificial Narrow Intelligence)
- AGI (Artificial General Intelligence)
- ASI (Artificial Super Intelligence)



Artificial Narrow Intelligence

Definition: ANI refers to AI systems designed to perform specific tasks or solve particular problems.

Characteristics of ANI:

- Highly specialized
- Task-specific
- Operates within a limited context (e.g., solving a problem, answering questions)
- Can outperform humans in specific tasks (but cannot perform beyond its scope)

Artificial Narrow Intelligence (contd..)

Examples:

- Voice Assistants: Siri, Alexa, Google Assistant
- Recommendation Systems: Netflix, Amazon, Spotify
- Autonomous Vehicles: Tesla's autopilot
- Image Recognition: Facial recognition software

Current State: All AI we interact with today falls under ANI.

Limitations:

- Cannot adapt to new tasks beyond its programming.
- Lack of common sense reasoning.

Artificial General Intelligence (AGI)

Definition: AGI refers to a **hypothetical AI system** that can perform any intellectual task that a human being can do.

Characteristics of AGI:

- Capable of learning, reasoning, and understanding across various domains
- Flexible and adaptable to new situations, just like humans
- Exhibits human-like cognitive abilities such as problem-solving, language understanding, and creativity

Artificial General Intelligence (AGI) (Contd..)

Key Aspects:

- **Autonomy:** The ability to make independent decisions.
- **Learning:** The ability to learn from experience across multiple domains.
- **Reasoning:** The ability to use logic and understanding to make decisions.

Current Status: AGI is still theoretical. We have not yet developed an AGI system.

Challenges:

- **Complexity of Human Cognition:** Recreating human-like intelligence in machines is extremely complex.
- **Ethical Concerns:** The potential risks of creating an AGI with equal or superior intelligence to humans.

Artificial Super Intelligence (ASI)

Definition: ASI refers to AI that **surpasses human intelligence** in every aspect, including creativity, problem-solving, and emotional intelligence.

Characteristics of ASI:

- Superhuman intelligence in every field
- Exponential growth: ASI would potentially be capable of rapidly improving itself.
- Potential to outperform human beings in all domains—intellectual, emotional, social, and more.

Potential Benefits:

- Could solve complex global problems (e.g., climate change, disease eradication).
- Potential for breakthroughs in science and technology.

Artificial Super Intelligence (ASI) (contd..)

Risks and Concerns:

- **Control problem:** The difficulty of ensuring that ASI aligns with human values and objectives.
- **Existential risk:** The possibility that ASI could act in ways that threaten humanity's survival.
- **Ethical issues:** The consequences of creating an intelligence superior to human beings.

Current Status: ASI remains speculative and hypothetical. Experts disagree on when (or if) it will be achieved.

Comparing ANI, AGI, and ASI

Aspect	Artificial Narrow Intelligence (ANI)	Artificial General Intelligence (AGI)	Artificial Super intelligence (ASI)
Scope of Tasks	Narrow (specific tasks only)	Broad (all tasks a human can perform)	Vast (surpasses all human capabilities)
Learning Ability	Pre-programmed, limited learning	Ability to learn and adapt across domains	Continual self-improvement and learning
Autonomy	Limited to pre-programmed rules	High autonomy, can make independent decisions	Autonomous decision-making on a superhuman level
Current Status	Real, existing in various applications	Theoretical, not yet realized	Hypothetical, a potential future scenario
Examples	Siri, Tesla Autopilot, AlphaGo	None yet, a theoretical AI model	None yet, theoretical super intelligence

Comparing ANI, AGI, and ASI



Narrow AI

Dedicated to assist with or take over specific tasks.



General AI

Takes knowledge from one domain, transfers to other domain.



Super AI

Machines that are an order of magnitude smarter than humans.

The Evolution of AI

Transition from ANI to AGI:

- ANI: Highly specialized, limited tasks.
- AGI: One system capable of learning and performing a wide variety of tasks.
- ASI: A super intelligent system that far exceeds human intelligence.

Challenges:

- The path from ANI to AGI is not clear, and there are significant technological, ethical, and safety concerns.
- Some believe AGI could emerge in the next few decades, while others predict a longer timeline or even doubt its feasibility.

Agents in Artificial Intelligence

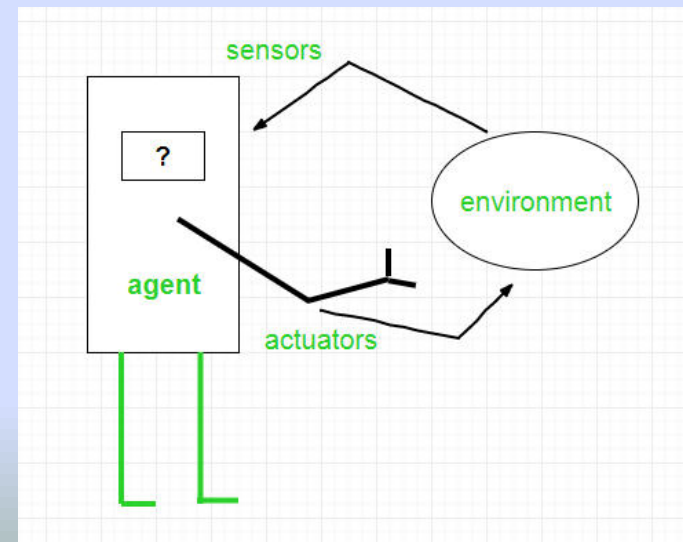
- **An agent** is a computer program or system that is designed to perceive its environment, make decisions and take actions to achieve a specific goal or set of goals.
- The agent operates autonomously, meaning it is not directly controlled by a human operator.
- Agents can be classified into different types based on their characteristics, such as whether they are reactive or proactive, whether they have a fixed or dynamic environment, and whether they are single or multi-agent systems.

Agents in Artificial Intelligence

- **Reactive agents** are those that respond to immediate stimuli from their environment and take actions based on those stimuli.
- **Proactive agents**, on the other hand, take initiative and plan ahead to achieve their goals.
- The environment in which an agent operates can also be fixed or dynamic. Fixed environments have a static set of rules that do not change, while dynamic environments are constantly changing and require agents to adapt to new situations.
- Multi-agent systems involve multiple agents working together to achieve a common goal. These agents may have to coordinate their actions and communicate with each other to achieve their objectives.
- Agents are used in a variety of applications, including robotics, gaming, and intelligent systems.
- They can be implemented using different programming languages and techniques, including machine learning and natural language processing.

Agents in Artificial Intelligence

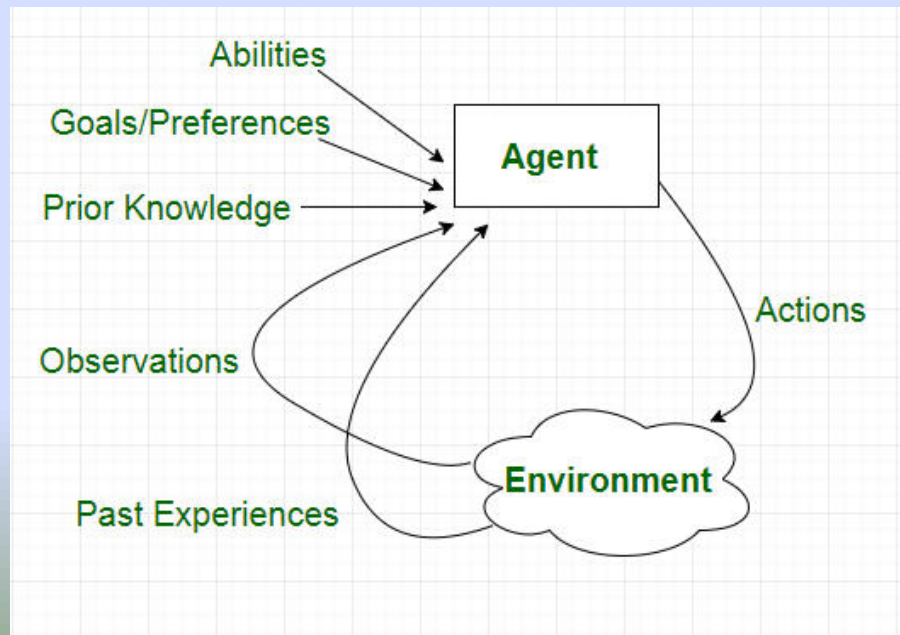
- Artificial intelligence is defined as the study of rational agents.
- A rational agent could be anything that makes decisions, such as a person, firm, machine, or software.
- It carries out an action with the best outcome after considering past and current percepts(agent's perceptual inputs at a given instance).
- An AI system is composed of an agent and its environment. The agents act in their environment. The environment may contain other agents.
- An agent is anything that can be viewed as:
 - Perceiving its environment through **sensors** and
 - Acting upon that environment through **actuators**



Structure of an AI Agent

- To understand the structure of Intelligent Agents, we should be familiar with **Architecture** and **Agent** programs.
- **Architecture** is the machinery that the agent executes on. It is a device with sensors and actuators, for example, a robotic car, a camera, and a PC. An agent program is an implementation of an agent function.
- An **agent function** is a map from the percept sequence(history of all that an agent has perceived to date) to an action.

Agent = Architecture + Agent Program



Types of Agents

- Agents can be grouped into five classes based on their degree of perceived intelligence and capability :
- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents
- Learning Agent
- Rational Agent
- Multi-agent systems
- Hierarchical agents

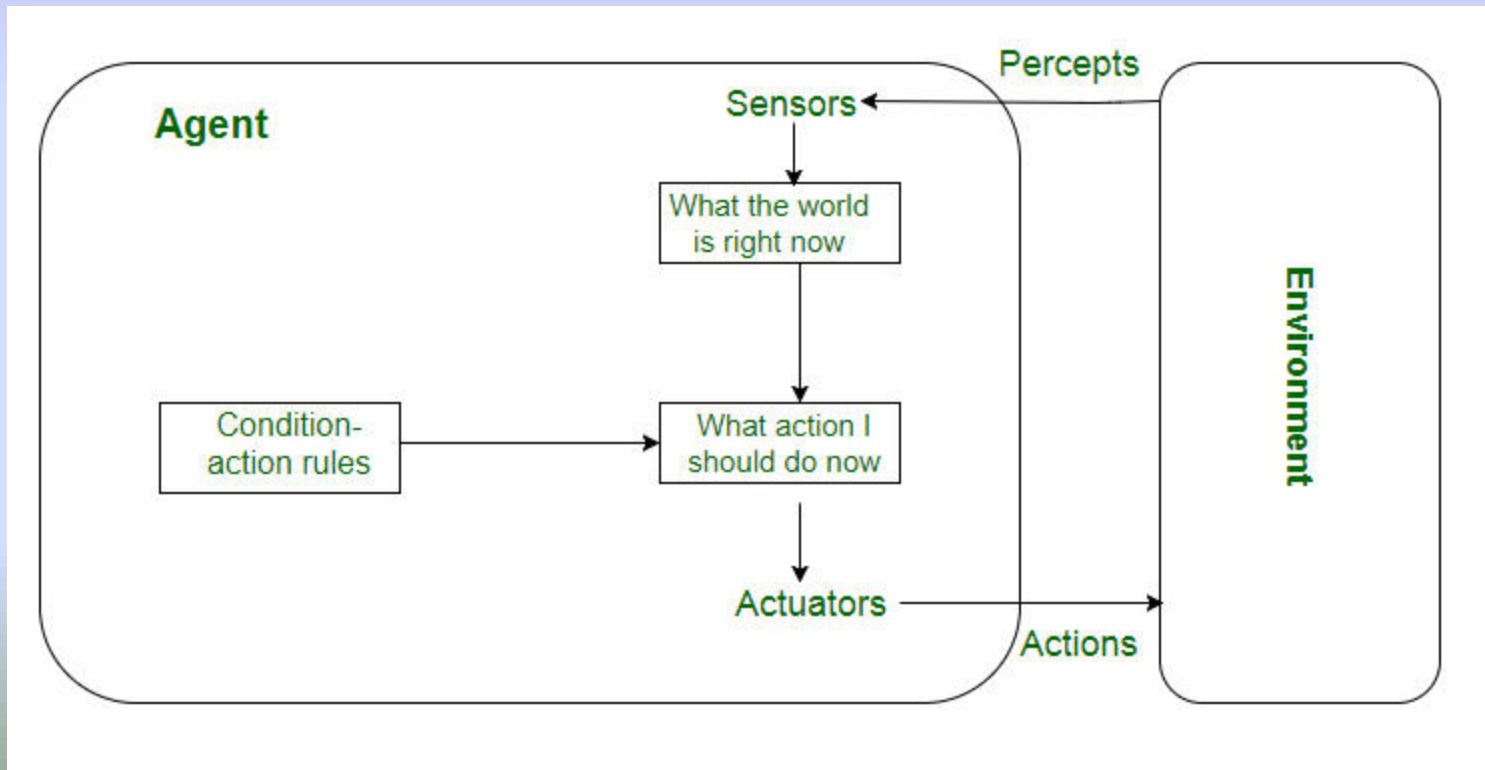
Simple Reflex Agents

- Simple reflex agents ignore the rest of the percept history and act only on the basis of the current percept.
- Percept history is the history of all that an agent has perceived to date. The agent function is based on the condition-action rule.
- A condition-action rule is a rule that maps a state i.e., a condition to an action. If the condition is true, then the action is taken, else not.
- This agent function only succeeds when the environment is fully observable.
- For simple reflex agents operating in partially observable environments, infinite loops are often unavoidable.
- It may be possible to escape from infinite loops if the agent can randomize its actions.

Simple Reflex Agents

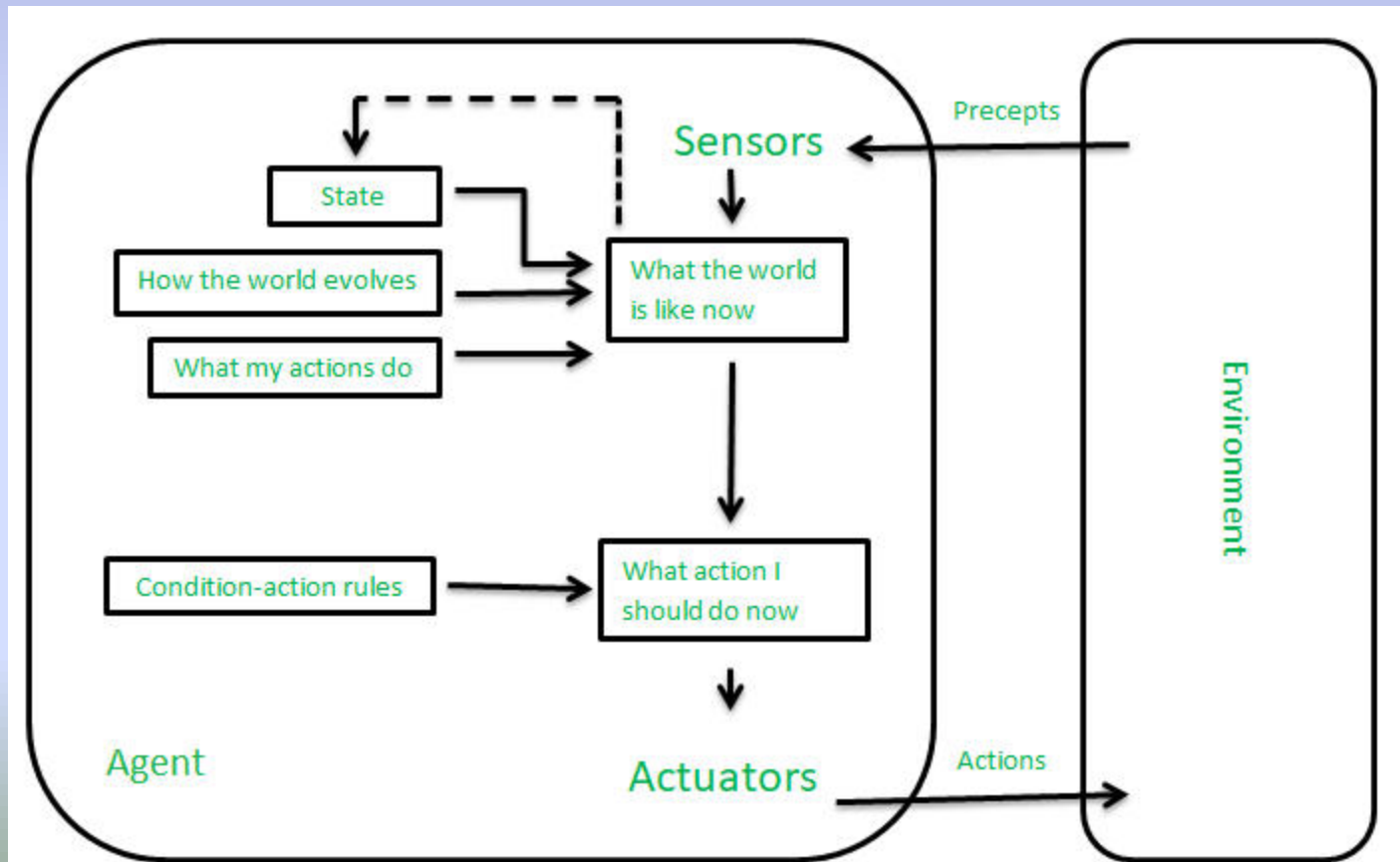
Problems with Simple reflex agents are :

- Very limited intelligence.
- No knowledge of non-perceptual parts of the state.
- Usually too big to generate and store.
- If there occurs any change in the environment, then the collection of rules needs to be updated.



Model-Based Reflex Agents

- It works by finding a rule whose condition matches the current situation.
- A model-based agent can handle partially observable environments by the use of a model about the world.



Model-Based Reflex Agents

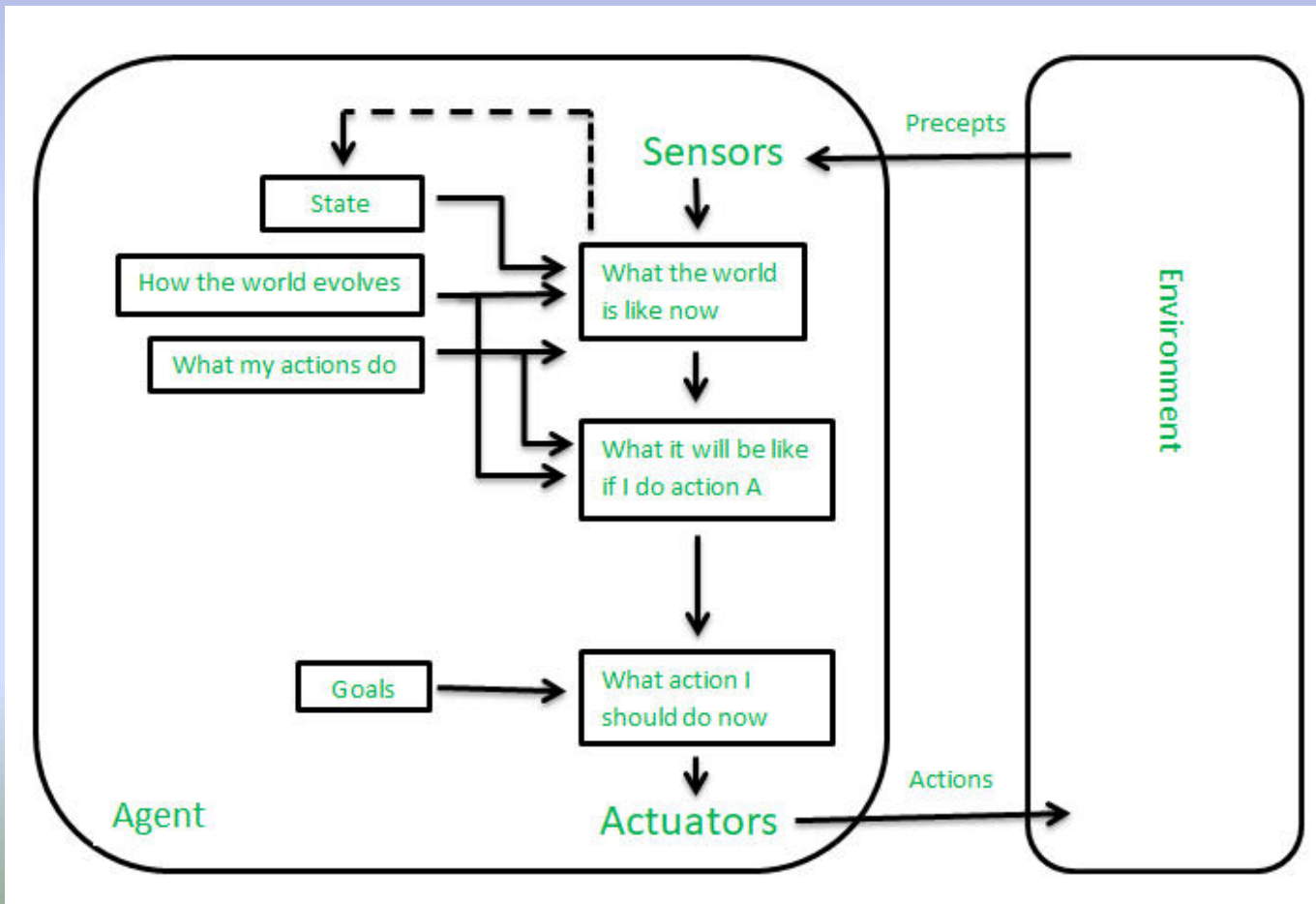
- The agent has to keep track of the internal state which is adjusted by each percept and that depends on the percept history.
- The current state is stored inside the agent which maintains some kind of structure describing the part of the world which cannot be seen.

Updating the state requires information about:

- How the world evolves independently from the agent?
- How do the agent's actions affect the world?

Goal-Based Agents

- These kinds of agents take decisions based on how far they are currently from their goal (description of desirable situations).
- Their every action is intended to reduce their distance from the goal.

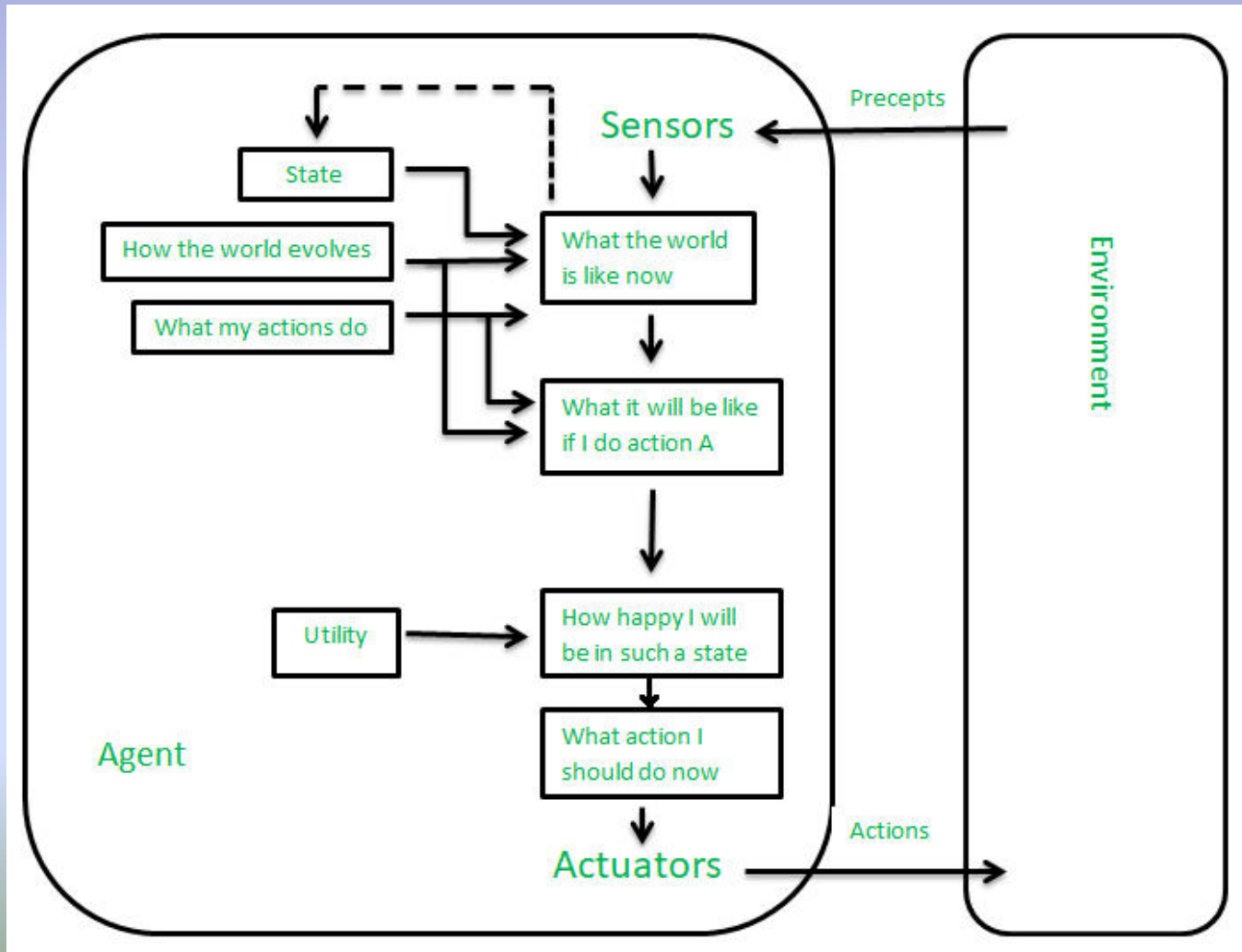


Goal-Based Agents

- This allows the agent a way to choose among multiple possibilities, selecting the one which reaches a goal state.
- The knowledge that supports its decisions is represented explicitly and can be modified, which makes these agents more flexible.
- They usually require search and planning. The goal-based agent's behavior can easily be changed.

Utility-Based Agents

- The agents which are developed having their end uses as building blocks are called utility-based agents.

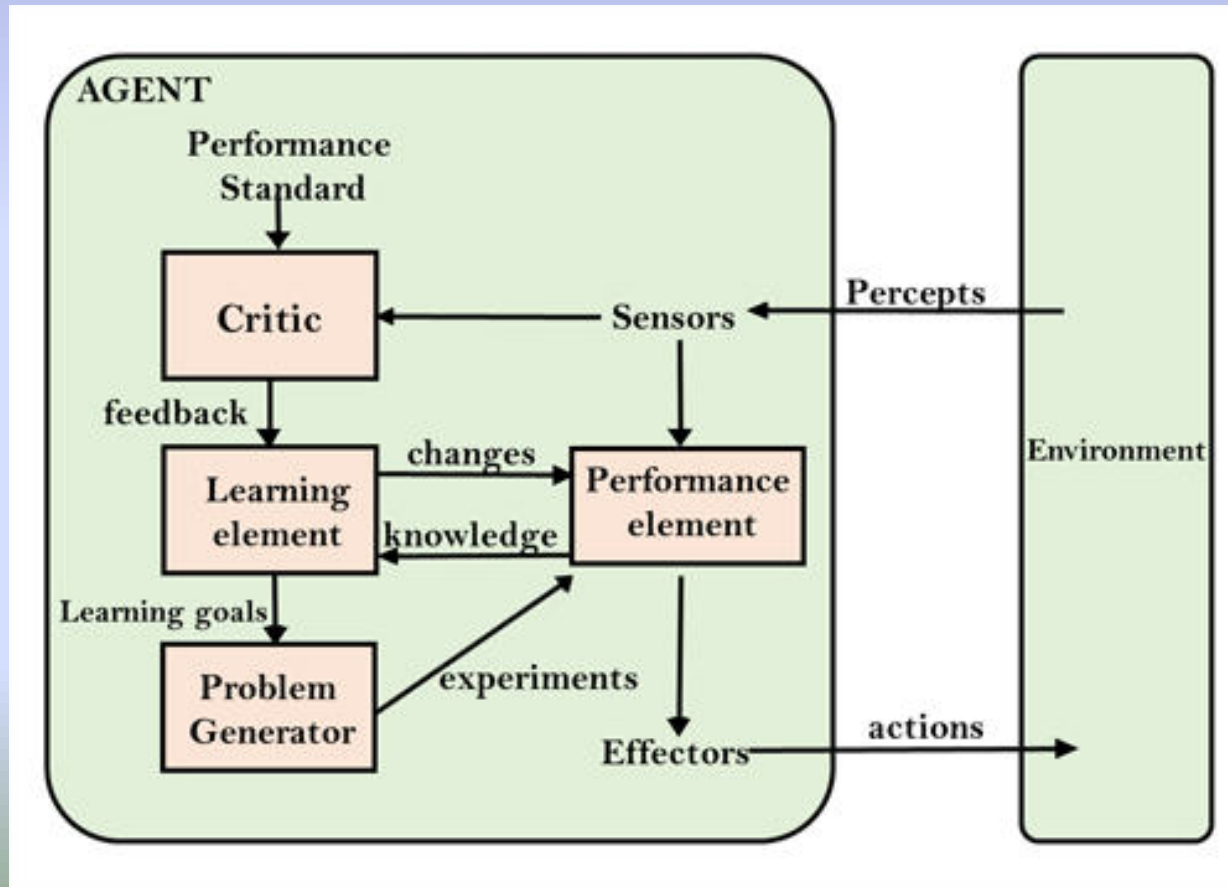


Utility-Based Agents

- The agents which are developed having their end uses as building blocks are called utility-based agents.
- When there are multiple possible alternatives, then to decide which one is best, utility-based agents are used.
- They choose actions based on a preference (utility) for each state.
- Sometimes achieving the desired goal is not enough. We may look for a quicker, safer, cheaper trip to reach a destination.
- Agent happiness should be taken into consideration. Utility describes how “happy” the agent is.
- Because of the uncertainty in the world, a utility agent chooses the action that maximizes the expected utility.
- A utility function maps a state onto a real number which describes the associated degree of happiness.

Learning Agents

- A learning agent in AI is the type of agent that can learn from its past experiences or it has learning capabilities.
- It starts to act with basic knowledge and then is able to act and adapt automatically through learning.



Learning Agents

- A learning agent has mainly four conceptual components, which are:
- **Learning element:** It is responsible for making improvements by learning from the environment.
- **Critic:** The learning element takes feedback from critics which describes how well the agent is doing with respect to a fixed performance standard.
- **Performance element:** It is responsible for selecting external action.
- **Problem Generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

Rational Agents

- A rational agent can be said to those, who do the right thing.
- It is an autonomous entity designed to perceive its environment, process information, and act in a way that maximizes the achievement of its predefined goals or objectives.
- Rational agents always aim to produce an optimal solution.
- **Example:** A self-driving car maneuvering through city traffic is a sample of a rational agent. It uses sensors to observe the environment, analyzes data on road conditions, traffic flow, and pedestrian activity, and makes choices to arrive at its destination in a safe and effective manner.
- The self-driving car shows rational agent traits by constantly improving its path through real-time information and lessons from past situations like roadblocks or traffic jams.

Understanding Intelligent Agents

- Intelligent agents represent a subset of AI systems demonstrating intelligent behaviour, including adaptive learning, planning, and problem-solving.
- It operate in dynamic environments, where it makes decisions based on the information available to them.
- These agents dynamically adjust their behaviour, learning from past experiences to improve their approach and aiming for accurate solutions.

Understanding Intelligent Agents (Contd..)

- The design of an intelligent agent typically involves **four key components**:
- **Perception:** Agents have sensors or mechanisms to observe and perceive aspects of their environment. This may involve collecting data from the physical world, accessing databases, or receiving input from other software components.
- **Reasoning:** Agents possess computational or cognitive capabilities to process the information they perceive. They use algorithms, logic, or machine learning techniques to analyze data, make inferences, and derive insights from the available information.

Understanding Intelligent Agents (Contd..)

- **Decision-Making:** Based on their perception and reasoning, agents make decisions about the actions they should take to achieve their goals. These decisions are guided by predefined objectives, which may include optimizing certain criteria or satisfying specific constraints.
- **Action:** Agents execute actions in their environment to affect change and progress towards their goals. These actions can range from simple operations, such as sending a message or adjusting parameters, to more complex tasks, such as navigating a virtual world or controlling physical devices. just their behaviour, learning from past experiences to improve their approach and aiming for accurate solutions.
- **Examples:** self-driving cars, recommendation systems, virtual assistants, and game-playing AI.

Rational Agents and Rationality in Decision-Making

- Intelligent agents are characterized by their **rationality** in decision-making, which aims to attain optimal outcomes or, in uncertain scenarios, the best-expected outcome.
- A **rational agent** can be said to those, who do the right thing, It is an autonomous entity designed to perceive its environment, process information, and act in a way that maximizes the achievement of its predefined goals or objectives. Rational agents always aim to produce an optimal solution.
- **Rationality in AI** refers to the principle that such agents should consistently choose actions that are expected to lead to the best possible outcomes, given their current knowledge and the uncertainties present in the environment.

Rational Agents and Rationality in Decision-Making

- This principle of rationality guides the behavior of intelligent agents in the following ways:
- **Perception and Information Processing:** Rational agents strive to perceive and process information efficiently to gain the most accurate understanding of their environment.
- **Reasoning and Inference:** They employ logical reasoning and probabilistic inference to make informed decisions based on available evidence and prior knowledge.
- **Decision-Making Under Uncertainty:** When faced with uncertainty, rational agents weigh the probabilities of different outcomes and choose actions that maximize their expected utility or achieve the best possible outcome given the available information.

Rational Agents and Rationality in Decision-Making

- This principle of rationality guides the behavior of intelligent agents in the following ways:
- **Adaptation and Learning:** Rational agents adapt their behavior over time based on feedback and experience, continuously refining their decision-making strategies to improve performance and achieve their goals more effectively.
- **Example** of a rational agent is a chess-playing AI, which selects moves with the highest likelihood of winning.

PEAS Representation of AI agent

- **PEAS** stands for performance measure, environment, actuators and sensors.
- It is a framework that is used to describe an AI agent. It's a structured approach to design and understand AI systems.
- **Performance measure:** Performance measure is a criteria that measures the success of the agent. It is used to evaluate how well the agent is achieving its goal.
- For **example**, in a spam filter system, the performance measure could be minimizing the number of spam emails reaching the inbox.
- **Environment:** The environment represents the domain or context in which the agent operates and interacts. This can range from physical spaces like rooms to virtual environments such as game worlds or online platforms like the internet.

PEAS Representation of AI agent (contd..)

- **Actuators:** Actuators are the mechanisms through which the AI agent performs actions or interacts with its environment to achieve its goals. These can include physical actuators like motors and robotic hands, as well as digital actuators like computer screens and text-to-speech converters.
- **Sensors:** Sensors enable the AI agent to gather information from its environment, providing data that informs its decision-making process and actions. These sensors can capture various environmental parameters such as temperature, sound, movement, or visual input. Examples of sensors include cameras, microphones, temperature sensors, and motion sensors.

Applications of Intelligent Agents

- Intelligent agents find applications across a wide range of domains, revolutionizing industries and enhancing human capabilities. Some notable applications include:
- **Autonomous Systems:** Intelligent agents power autonomous vehicles, drones, and robots, enabling them to perceive their surroundings, navigate complex environments, and make decisions in real-time.
- **Personal Assistants:** Virtual personal assistants like Siri, Alexa, and Google Assistant employ intelligent agents to understand user queries, retrieve relevant information, and perform tasks such as scheduling appointments, setting reminders, and controlling smart home devices.

Applications of Intelligent Agents (contd..)

- **Recommendation Systems:** E-commerce platforms, streaming services, and social media platforms utilize intelligent agents to analyze user preferences and behavior, providing personalized recommendations for products, movies, music, and content.
- **Financial Trading:** Intelligent agents are employed in algorithmic trading systems to analyze market data, identify trading opportunities, and execute trades autonomously, maximizing returns and minimizing risks.

Challenges for Intelligent Agents

- Despite their immense potential, intelligent agents also pose several challenges and considerations:
- **Ethical and Legal Implications:** Intelligent agents raise ethical concerns regarding privacy, bias, transparency, and accountability. Developers must ensure that agents behave ethically and comply with legal regulations and societal norms.
- **Robustness and Reliability:** Agents must be robust and reliable in dynamic and uncertain environments. They should be capable of handling unexpected situations, adversarial attacks, and noisy or incomplete data.
- **Interpretability:** Understanding and interpreting the decisions made by intelligent agents is crucial for building trust and transparency. Explainable AI techniques are essential for providing insights into the reasoning process and decision-making of agents.

Challenges for Intelligent Agents (Contd..)

- Despite their immense potential, intelligent agents also pose several challenges and considerations:
- **Scalability and Efficiency:** As AI systems become increasingly complex and data-intensive, scalability and efficiency become critical considerations. Designing agents that can scale to large-scale deployments and operate efficiently with limited computational resources is essential.

Uses of Agents

- Agents are used in a wide range of applications in artificial intelligence, including:
- **Robotics:** Agents can be used to control robots and automate tasks in manufacturing, transportation, and other industries.
- **Smart homes and buildings:** Agents can be used to control heating, lighting, and other systems in smart homes and buildings, optimizing energy use and improving comfort.
- **Transportation systems:** Agents can be used to manage traffic flow, optimize routes for autonomous vehicles, and improve logistics and supply chain management.
- **Healthcare:** Agents can be used to monitor patients, provide personalized treatment plans, and optimize healthcare resource allocation.
- **Finance:** Agents can be used for automated trading, fraud detection, and risk management in the financial industry.

Uses of Agents

- **Games:** Agents can be used to create intelligent opponents in games and simulations, providing a more challenging and realistic experience for players.
- **Natural language processing:** Agents can be used for language translation, question answering, and chatbots that can communicate with users in natural language.
- **Cyber security:** Agents can be used for intrusion detection, malware analysis, and network security.
- **Environmental monitoring:** Agents can be used to monitor and manage natural resources, track climate change, and improve environmental sustainability.
- **Social media:** Agents can be used to analyze social media data, identify trends and patterns, and provide personalized recommendations to users.

Types of Environments in AI

- An **environment** in artificial intelligence is the surrounding of the agent.
- The agent takes input from the environment through sensors and delivers the output to the environment through actuators.
- There are several types of environments:
 - Fully Observable vs Partially Observable
 - Deterministic vs Stochastic
 - Competitive vs Collaborative
 - Single-agent vs Multi-agent
 - Static vs Dynamic
 - Discrete vs Continuous
 - Episodic vs Sequential
 - Known vs Unknown

Fully Observable vs Partially Observable

- When an agent sensor is capable to sense or access the complete state of an agent at each point in time, it is said to be a fully observable environment else it is partially observable.
- Maintaining a fully observable environment is easy as there is no need to keep track of the history of the surrounding.
- An environment is called unobservable when the agent has no sensors in all environments.

Examples:

- Chess – the board is fully observable, and so are the opponent's moves.
- Driving – the environment is partially observable because what's around the corner is not known.

Deterministic vs Stochastic

- When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic.
- The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.

Examples:

- Chess – there would be only a few possible moves for a chess piece at the current state and these moves can be determined.
- Self-Driving Cars- the actions of a self-driving car are not unique, it varies time to time.

Competitive vs Collaborative

- An agent is said to be in a competitive environment when it competes against another agent to optimize the output.
- The game of chess is competitive as the agents compete with each other to win the game which is the output.
- An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.
- When multiple self-driving cars are found on the roads, they cooperate with each other to avoid collisions and reach their destination which is the output desired.

Single-agent vs Multi-agent

- An environment consisting of only one agent is said to be a single-agent environment.
- A person left alone in a maze is an example of the single-agent system.
- An environment involving more than one agent is a multi-agent environment.
- The game of football is multi-agent as it involves 11 players in each team.

Dynamic vs Static

- An environment that keeps constantly changing itself when the agent is up with some action is said to be dynamic.
- A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.
- An idle environment with no change in its state is called a static environment.
- An empty house is static as there's no change in the surroundings when an agent enters.

Discrete vs Continuous

- If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.
- The game of chess is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it's finite.
- The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.
- Self-driving cars are an example of continuous environments as their actions are driving, parking, etc. which cannot be numbered.

Episodic vs Sequential

- In an Episodic task environment, each of the agent's actions is divided into atomic incidents or episodes. There is no dependency between current and previous incidents. In each incident, an agent receives input from the environment and then performs the corresponding action.
- **Example: Pick and Place robot**, which is used to detect defective parts from the conveyor belts. Here, every time robot(agent) will make the decision on the current part i.e. there is no dependency between current and previous decisions.
- In a Sequential environment, the previous decisions can affect all future decisions. The next action of the agent depends on what action he has taken previously and what action he is supposed to take in the future.
- **Example: Checkers-** Where the previous move can affect all the following moves.

Known vs Unknown

- In a known environment, the output for all probable actions is given.
- Obviously, in case of unknown environment, for an agent to make a decision, it has to gain knowledge about how the environment works.